threshold for the query evaluation score to avoid training on such false-positives. For future work, we aim to explore more sophisticated approaches for query evaluation.

5.2 Contrastive Preference Optimization

One key assumption we made throughout this paper was that using better generator models will translate into higher quality of downstream performance for our reranker using synthetic data. However, this does not hold true in general, as seen in rows (3.1) and (3.2) in Table 2.

We determine that further research is essential to determine the potential impact of optimizing synthetic data generator models for information retrieval. In particular, we plan to investigate better ways to generate training datasets to encourage more appropriate query generation.

5.3 CoT Prompts

As shown in Table 2, incorporating a guided process improves query quality; however, this comes with trade-offs and requires careful consideration of certain aspects.

While we anticipated that applying CoT would improve query quality and reduce the number of instances required for training, this was not consistently observed, particularly for the TREC-Covid dataset in figure 5. This could be due to several factors:

- The TREC-Covid dataset represents a corpus of 171k documents. Training on a subset of the highest scoring documents seems to decrease the performance of the reranker. Training on higher number of samples, instead of filtering, seems to provide better training signal translated into better downstream performance.
- Although CoT reasoning introduces guided steps for query generation, this can sometimes result in diminished performance. CoT prompts may lead the model to generalize in a way that detracts from the relevance of the generated queries. For instance, in some cases, the reasoning steps cause the model to focus on peripheral concepts rather than extracting keywords or core topics directly relevant to the document.

These results, however, relate to the specific the datasets we test on. Further investigation across a larger pool of corpora is still needed. To answer then the question whether a filtering strategy is needed, the answer seems not evident as of now. Nonetheless, we note that using dynamic prompts instead of a static hardcoded template does improves the downstream performance of the reranker model, as indicated in the top scores for the 4 datasets shown.

6 Conclusion

In this study, we reproduce and verify the main claims of InPars Toolkit, confirming its claims of providing an end-to-end reproducible pipeline and a clear guide for reproducing InPars-V1, InPars-V2, and partially Promptagator. We also validate the pipeline's plug-and-play capability by integrating LLaMA 3.1 8B as the generator model. Our results closely match the ones reported by the original authors, supporting the toolkit's reliability.

Beyond reproduction, we extend the toolkit in two ways. First, we apply Contrastive Preference Optimization (CPO) to fine-tune the generator LLM to guide the model into generating generally better queries. Second, we apply CoT for dynamic prompt optimization in lieu of fixed templates. This generally improved query quality, though some cases revealed complexities which deserve further investigation.

Overall, our reproductions and extension experiments build upon the versatility of the InPars Toolkit and highlight practical pathways for future research.

Table 4: Sample document from Scifact with its reference query (finding) and queries from the generator models.

	Text	Score
Document	Myelodysplastic syndromes (MDS) are age-dependent stem cell malignancies that...	100
Reference *Finding*	Toll-like receptor (TLR) signaling is involved in the pathogenesis of human MDS.	36.1
Teacher *Finding*	Myeloid-derived suppressor cells (MDSC) are... hematopoiesis. MSDC expansion...	96.3
Student *Finding*	Myeloid-derived suppressor cells (MDSC) are ... of ineffective hematopoiesis.	94.2
CPO *Finding*	Myelodysplastic syndromes (MDS) are age-dependent stem cell malignancies that...	**99.6**

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